

OSSP:

Commands:

uname – Displays operating system and system details.

lscpu – Displays CPU architecture and processor information.

```
LST@LAPTOP-EK5DCJVE:~# gcc teja.c -o teja
LST@LAPTOP-EK5DCJVE:~# ./teja
Enter a Linux command: uname

Parent Process
Parent PID: 624
Waiting for child process...

Child Process
Child PID: 629
Linux
Child process completed.
```

ps – Shows the list of currently running processes.

```
LST@LAPTOP-EK5DCJVE:~# ./teja
Enter a Linux command: ps

Parent Process
Parent PID: 673
Waiting for child process...

Child Process
Child PID: 674
  PID TTY          TIME CMD
  323 pts/0        00:00:00 bash
  673 pts/0        00:00:00 teja
  674 pts/0        00:00:00 ps
Child process completed.
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system manages CPU usage by allocating processor time to each program. This allows multiple applications to run smoothly at the same time.

Example: Chrome, VS Code, and a music player can run together without interrupting each other.

2. Memory (RAM) → Memory Allocation

The operating system assigns separate memory space to every running program and prevents memory conflicts.

Example: A browser and a text editor each use their own RAM while running.

3. Storage (SSD/HDD) → File Management

The operating system organizes files and folders, making it easy to store, retrieve, and manage data.

Example: Saving a PDF or image in the Downloads folder.

4. I/O Devices → Device Control

The operating system communicates with input and output devices through device drivers, ensuring they work correctly.